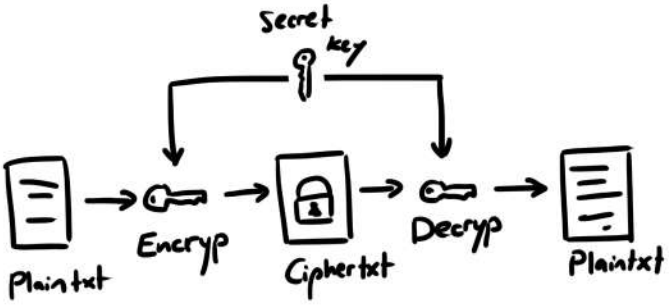
Chapter 2

Symmetric Encryption



- disadvg: key exchange
- advg: very fast, implementable in HW

→ **Block Cipher** advanced encryption standard AES

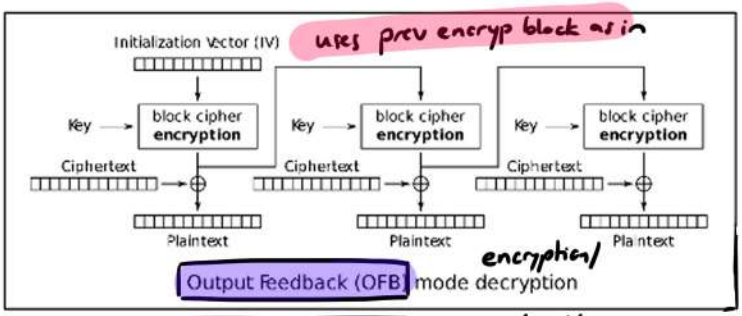
- process plain and ciphertxt in blocks (e.g. 128bits)

→ **Padding** ISO Padding

irregular but reconstructable

- fill up incomplete block with "regular" pattern

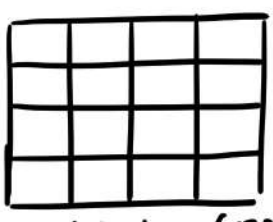
Block Cipher Operation Modes



Electronic Code Book (ECB)

- txt block fixed size + key = block of same size encrypted
- uses same key for all
- can encryp in parallel

AES Algorithm



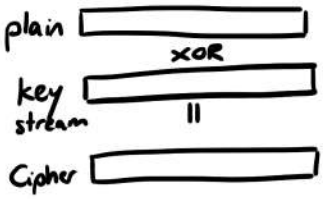
4x4 bytes (128bits)
→ 10 Rounds

Round of Operations:

1. SubBytes (substitute bytes via lookup table)
2. ShiftRows (permuting particular num of times)
3. MixColumns (matrix mult, each col mult matrix)
4. AddRoundKey (XOR with corresponding round key)

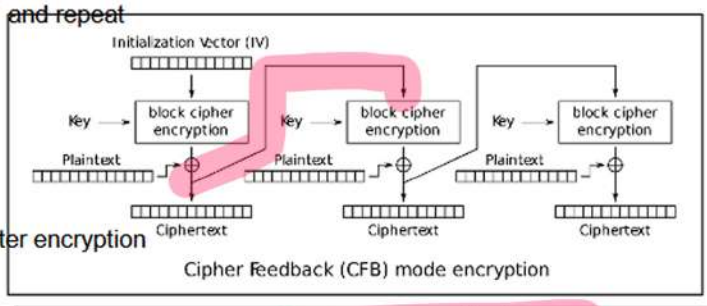
→ **Stream Cipher** ChaCha20

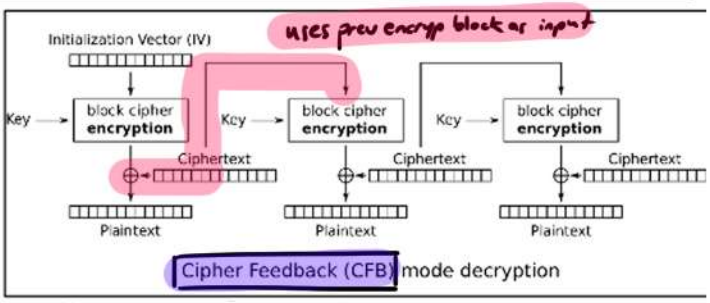
- key of fixed length
- generate data stream of arbitrary length
- convert each bit of plain to cipher using XOR



Variants

- Synchronous stream ciphering
  → gen key stream independently of plain or key txt
- self-synchronizing stream ciphering
  → depends on prev encryp bit

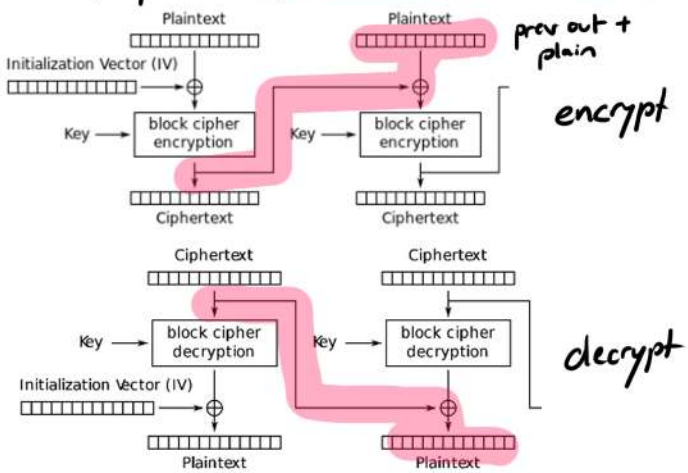




better than

Cipher Block Chaining (CBC)

- encryp of block depends on predecessor blocks



Init Vector

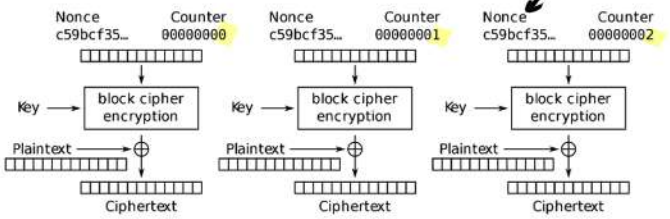
→ must not be secret
→ should be random and different for each msg
→ same size as block (AES 128 bit)

errors:
- use key as IV
- set IV to 0 or other fixed val
- gen IV from pswd

→ not random and possibly exposed

Counter Block Mode (CBM)

number used only once



Counter (CTR) mode encryption

$C_i = P_i \oplus E_k(ctr_i)$

- each block new unpredictable keystream

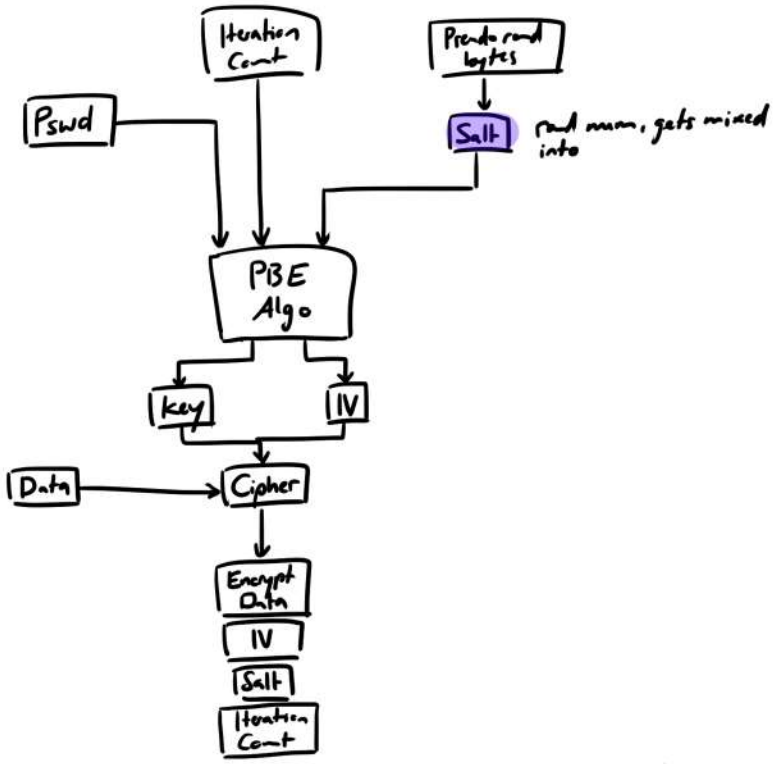
adv: key does not depend on prev → parallelisation
key stream can be precomputed
errors only affect one block

Password Based Encryption (PBE)

- gen cryptographic keys from pswd

PKCS#5 using PBKDF2 → pswd based key derivation function 2

PKCS#12 → file format that stores private keys with associated certificate in pswd-protected manner



But: case-by-case effectiveness
- ciphertxt can be modified and influences plain
- Padding Oracle Attack
- always use CBC, CTR, etc with authentication

→ Secure encryption with authentication!

Authenticated Encryption

- authenticated if not only confidentiality but also integrity of data to be encrypted is given

Galois Counter Mode

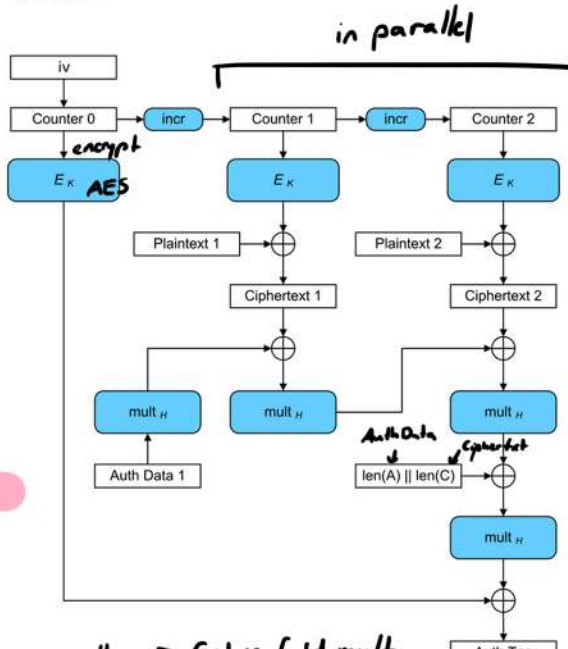
AES with Counter mode for encrypt and Galois mode for authentication

↳ Counter Mode: transforms block into stream cipher
(block: fixed size; stream: continuous bit by bit)
by encrypting successive values of a counter
→ Counter + secrKey = keystream
↳ keystream ⊕ plaintxt = ciphertxt

↳ Galois Mode: msg auth by gen a tag that verifies authenticity of data

↓
output is ciphertxt and authentication tag

parallel processing into ciphertxt possible



$mult_H$ → Galois field mult like hash func

Authenticated Encryption With Associated Data — AEAD

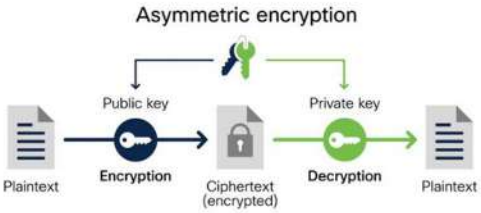
- provides encryption and authentication + associate additional data to encrypt msg
→ associated data prevents replay attacks by providing authenticity to ciphertxt
↳ makes modifications detectable
- associated data can be e.g. addresses, ports sequence nums, etc.
→ e.g. address and seqNum as AD, it now won't work for different address and seqN

Base 64 Encoding

- for transmission of binary data in seq of 8-bit bytes across channels that only reliably support txt content
- 64 printable ASCII chars
→ Principle:
split 24 bits into 4x6 bits
extend each 6 bits → 8 bits
→ disadv.: msg 33% longer

Asymmetric Encryption

- private and public key
→ disadv.: high computing effort
→ adv.: key exchange no prob



→ Rivest-Shamir-Adleman (RSA)

- based on difficulty of factoring large nums to hard against attack
- currently: key length of 3000
- much slower than AES

algo:
1. two large prime num $> 2^{512}$
2. mult primes p q
3. calc Euler's Totient Function
$\phi(n) = (p-1)(q-1)$
4. pub exponent e → $1 < e < \phi(n)$
5. calc priv exponent
$d \times e = 1 \pmod{\phi(n)}$
→ solve for d

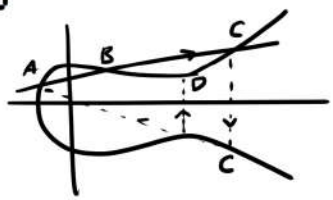
greatest common divisor $\gcd(e, \phi(n)) = 1$
commonly chosen as small prime (e.g. 65537)
not divide $\phi(n)$

→ PubKey (n, e)
PrivKey (n, d)

encrypt: $c = m^e \bmod n$
decrypt: $m = c^d \bmod n$

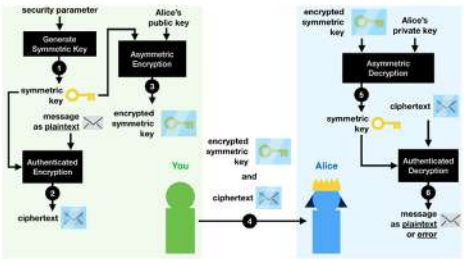
Elliptic Curve Cryptography

- several passes with straight line through two points and x-axis reflection of intersection
→ PubKey: start and endpoint
PrivKey: num rounds



Hybrid Encryption

- sym and asym encrypt
- msg encrypted sym
sym key encrypted asym
→ adv.: high speed encrypt
secure key exchange



2.5 Practical Aspects

✕ Security by Obscurity → i.e. parts of encryption process must be kept secret to make it secure

○ Open Design → sec does not depend on secrecy of its design or implementation

→ always use crypto libraries, never own

Compression - Encryption - Error Detection
↳ makes encrypt faster and cryptanalysis harder exploit of redundancy in txt
↳ add. signature/checksum → manipulations recognisable

Key lengths → sym: ≥ 256, atk via brute force
asym: ≥ 3000 (RSA), 250 (ECC), math methods

Formats for storing encrypt

- XML encrypt standard
- PKCS#7 enveloped data

Attacks

- Cryptanalysis: mathematical, statistical or implementation weaknesses
- Brute Force
- Social Engineering
- obtain keys via vulnerabilities in sys
- Side Channel Attack via phys measurements of sys (e.g. energy consumption, time consump)
- Fault Attack: causes faulty exec of crypto operation
- Man in the Middle Attack

Key Management

→ generation, storage of keys
withdrawal of invalid or compromised keys
sec transmission / exchange of keys
backup copies
destruction of keys

Key Generation
- good rand nums req
- gen in secure place

- key compromised → lock - exchange - recrypt
  - lock → key
  - recrypt → all data
- secure transmission / exchange of keys, mostly based on asym cryptography

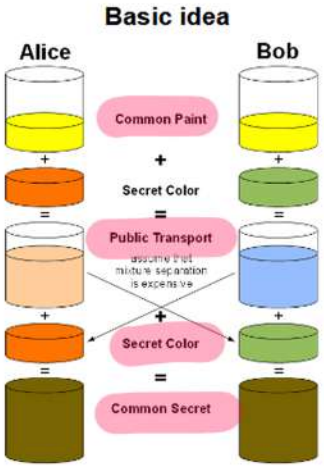
Storage

- in config files, environ vars with strict access control or Password Based Encryption
- external keystores: Vaults / Secret Manager
  ↳ access via API with authe, authz and audit logs
- chip card with pin
- key splitting (n : m principle)
  - n: secrets needed to reconstruct key
  - m: secrets
  - n > m

Exchange

Diffie - Hellman

- pub key algo for dist of keys
- not suitable for encrypt
- determining final secret even with col from pub trans is computationally expensive and impossible to compute in practical time



Elliptic Curve Diffie Hellman

- EC sys params are pub

p	prime number
a,b	curve parameter
P	base point on curve
q	order of P
i	cofactor

Alice: priv key $x \in [1, \dots, q-1]$, sends $Q_A = xP$

Bob: sends same with priv key y

→ same as DH, both calc priv key · $Q_{A/B}$ for common secret

elliptical curve discrete logarithm problem (ECDLP)

→ computat. infeasible to find k where $Q = k \cdot P$
- Q: target
- P: base point

Save / Destroy

- backup copies necessary → exceptional situation
  - → Key Recovery (in case of key loss)
  - → Key Escrow (for law enforcement)
  - → Master Key

where to store and who can access it?

Destruct:
- overwrite files and hard disk mem areas
  delete cache and swap areas
- possibly destroy hardware